SSTG-Nav: Metric-Grounded Spatial-Semantic Topological Graphs for Reusable Object Navigation

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Abstract

*Robots operating in familiar environments should reuse prior observations to guide subsequent object navigation. Recognizing a remembered object, however, does not specify a reachable approach pose, and observations of one instance can support distinct approach sides. We present **SSTG-Nav**, a metric-grounded **S**patial-**S**emantic **T**opological **G**raph framework that couples reusable object identity with calibrated approach geometry for **Nav**igation. A single goal-independent RGB-D survey stores calibrated observations and their connectivity. Category, description, and photograph queries select spatial instances from this shared memory; depth and source poses supply ordered object-centric approaches. Source-aware consolidation compresses the semantic index, while a linked residual bank retains alternative approach geometry. On GOAT-Bench, the shared survey supports complete multimodal sequences and achieves 64.5% Unseen success under the strict 0.25 m viewpoint criterion with navmesh-assisted execution. Matched HM3D experiments hold detections and rankings fixed and isolate a 15.3-percentage-point success gain from replacing observation poses with metric standoffs. Online MultiON evaluation demonstrates reuse of observations acquired during navigation, while repeated-request studies quantify survey cost amortization. These results establish measured approach geometry as a key component of reusable semantic memory for repeated object navigation.*

1. Introduction

A robot returning to the same home should use its previous observations when asked to find a chair, a particular bed, or an object described by its surroundings. Object navigation has advanced through semantic exploration, vision-language reasoning, and learned spatial memory [3, 31–33]. Sequential and lifelong benchmarks make persistent experience increasingly valuable [2, 9, 28]. In this setting, the central question extends beyond finding a target: what should a robot retain so that later requests become executable navigation goals?

Semantic identity and approach geometry answer different parts of this question. A robot may recognize a bed through a doorway while its observation pose remains in the corridor, too far away to complete a later request. Observations from different sides may identify the same bed while supporting different approaches. A reusable memory must associate these observations without discarding the geometry needed to reach the selected instance. Storing a semantic label at each observation pose leaves this *observation-to-destination gap* unresolved. Merging all repeated detections into one destination reduces redundancy but removes alternative approach sides that can remain accessible when another is obstructed.

We introduce SSTG-Nav, linking spatial-semantic instances to source-supported metric approaches (Figure 1). Category, description, and photograph queries select evidence from a shared goal-independent RGB-D survey. Depth and source poses turn that evidence into metric standoffs, grouped by instance to preserve semantic consistency across approach sides. Source-aware consolidation retains a compact primary index, while a linked residual bank preserves alternative geometry.

Our main contributions are summarized as follows:

- A metric-grounded spatial-semantic topological memory linking object identity, calibrated source observations, and measured approach alternatives. A source-aware primary index and linked residual bank separate retrieval compactness from retained geometric diversity.
- A multimodal query interface that resolves category, description, and photograph goals against one shared survey. Complete instance-specific approach sequences preserve semantic priority and alternative approach sides across successive requests.

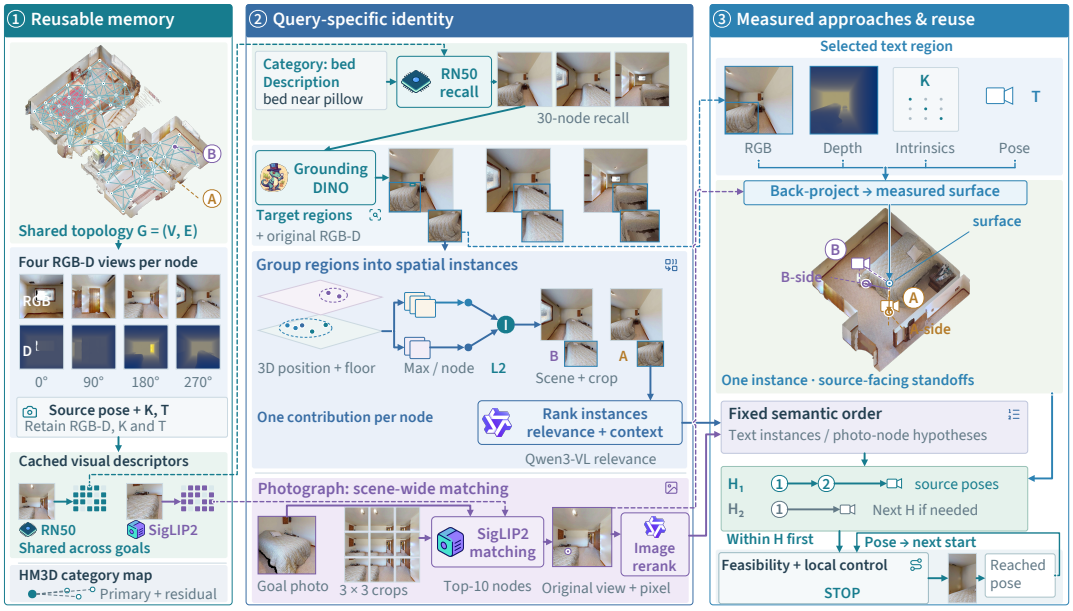


Figure 1. **Overview of SSTG-Nav. (1) Reusable memory.** The goal-independent survey retains calibrated RGB-D views and shared descriptors. The upper scene shows recorded RGB-D geometry with survey topology and source nodes A/B; the HM3D inset links primary and residual memory. **(2) Query-specific identity.** Category and description queries ground regions, form source-aware instances, and rank them by relevance and context. Photograph queries independently match and rerank scene-wide views. **(3) Measured approaches and reuse.** Both branches back-project selected pixels to measured surfaces and derive source-facing standoffs using RGB-D calibration. The bedroom cutaway provides simulator context: A/B support different approach sides of one bed. Execution tries each hypothesis's standoffs and source-pose fallbacks in fixed H_1, H_2 order; the reached pose starts the next query.

- Benchmark evaluation on GOAT-Bench and online MultiON, with matched HM3D studies that isolate endpoint geometry and quantify storage and acquisition costs. A physical-system demonstration connects calibrated perception, persistent memory, and robot execution.

2. Related Work

Open-vocabulary navigation. Object navigation uses learned spatial context [3, 23] and pretrained vision-language representations [5, 19] to guide exploration toward a target. ESC introduces commonsense constraints [33], VLFM scores exploration frontiers [32], and SG-Nav reasons over an online scene graph [31]. SSTG-Nav focuses on reusing acquired evidence across navigation requests, linking remembered object identity to measured approaches for subsequent goals.

Reusable multimodal scene memory. VLMaps, ConceptFusion, and CLIP-Fields support language-grounded spatial queries [7, 8, 26], while reusable abstract models accumulate knowledge for later ObjectNav tasks [1]. GOAT retains category and multi-view instance appearance [2]; GOAT-Bench evaluates sequences of category, description, and photograph goals [9]. TANGO composes navigation primitives [35], AstraNav-Memory compresses visual histories [6], and MTU3D jointly learns grounding and frontier selection [34]. SSTG-Nav couples multimodal instance selection with source-supported approach alternatives, preserving the link between a remembered object's identity and its distinct approach sides.

Geometry and topological memory. TSGM and TopoNav organize navigation with semantic graphs [11, 14], while SGoLAM and MOPA retain mapped goals for sequential navigation [10, 24]. CARe addresses retrieval uncertainty in pre-explored semantic maps through confidence and multi-view consistency [12]. APRR actively refines final position and orientation for manipulation [21], highlighting the role of terminal geometry in task completion. SSTG-Nav retains multiple source-supported standoffs alongside a compact semantic index, preserving alternative approach geometry when repeated observations are consolidated.

3. Method

SSTG-Nav separates semantic instance selection from approach construction while linking both to their source observations. Its persistent state comprises a survey graph $\mathcal{G}$, calibrated RGB-D observations, and linked instance memory

$\mathcal{H}$. A category, description, or photograph request ranks instance hypotheses from modality-specific evidence. Each hypothesis carries an ordered list of measured approaches, which the controller executes from the current pose. The reached pose becomes the next request's start, while acquired evidence persists. Figure 1 summarizes this pipeline.

3.1. Reusable Scene Memory

A goal-independent survey constructs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ within each connected navigable component. Node $v_i = (\mathbf{p}_i, \mathbf{q}_i, \mathcal{I}_i)$ stores the agent's base position, orientation, and four calibrated cardinal RGB-D views. These views cover horizontal headings at a fixed per-node storage cost. Greedy farthest-point sampling over a fixed navigable-point pool provides spatial coverage independently of target categories. Nearby nodes connect when a local navmesh path exists, weighted by geodesic length. Future requests, query starts, object annotations, and valid goal viewpoints remain outside survey construction. The graph organizes observations and a continuous survey walk; the platform planner executes query routes.

Whole-view and local-crop descriptors are computed once and attached to the original observations. This survey bank supplies shared visual evidence across modalities. Query-conditioned detections subsequently associate the relevant regions into instances; keeping the original RGB-D and calibration lets different requests reuse the same source geometry without rebuilding the survey.

A text detection $d = (c_d, b_d, q_d, k_d)$ at node i records its semantic label, bounding box, confidence, and source view. A normalized pixel $\tilde{\mathbf{u}}_d$ localizes each detection or photograph hypothesis: the box center for text and an appearance-weighted location for photographs. Forward depth z_d is the median of valid values in $(0.2, 6.0)$ m within a 7×7 patch. With normalized-image intrinsics K, camera rotation R_{ik_d}, and optical-center translation $\mathbf{t}_{ik_d}$, the surface estimate is

$$\hat{\mathbf{o}}_d = R_{ik_d} z_d S K^{-1} [\tilde{\mathbf{u}}_d^\top, 1]^\top + \mathbf{t}_{ik_d}, \tag{1}$$

where $S = \mathrm{diag}(1, -1, -1)$ converts image-down, forward-positive rays to Habitat's camera axes. The translation includes the sensor's height above the agent base; Supplementary Sec. A.2 specifies calibration and invalid-depth handling. An observation through a doorway can recognize an object while leaving the camera outside its approach region. We compute a source-facing standoff from the measured surface (Figure 2):

$$\begin{aligned} \Pi(\mathbf{s}_d^*) &= \Pi(\hat{\mathbf{o}}_d) + \rho \frac{\Pi(\mathbf{p}_i - \hat{\mathbf{o}}_d)}{\|\Pi(\mathbf{p}_i - \hat{\mathbf{o}}_d)\|_2}, \\ (\mathbf{s}_d^*)_y &= (\mathbf{p}_i)_y. \end{aligned} \tag{2}$$

where Π projects onto the horizontal plane, y is vertical, and $\rho = 0.8\,\mathrm{m}$. Setting the source base height preserves the observed floor. For GOAT text instances, the effective offset is $\min(\rho, \|\Pi(\mathbf{p}_i - \hat{\mathbf{o}}_d)\|_2)$: a closer source retains its own pose. Navigable snapping supplies executable alternatives (Sec. 3.4). Each approach remains linked to its original RGB region, depth, and source pose.

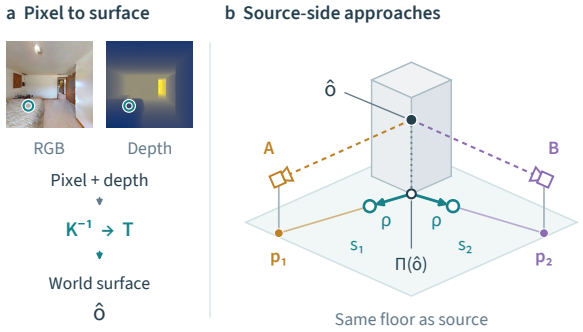


Figure 2. **Source-supported approach geometry.** (a) Calibrated RGB-D maps a selected pixel to a visible surface estimate. (b) Schematic source-facing offsets ρ lie on the source agent's floor, preserving alternative approach sides. Dashed rays show observation support; solid arrows show offsets. Navigable snapping precedes execution.

3.2. Source-Aware Instances and Residual Approaches

Repeated detections at one pose provide redundant semantic support, whereas different observation poses can expose distinct approach sides. The category-map instantiation used for controlled HM3D studies retains standoffs reachable from their source. It associates category-consistent detections by horizontal and vertical proximity and short geodesic connectivity, keeping nearby but disconnected access regions separate. Within cluster C, each observing node contributes its strongest confidence:

$$Q(C) = 1 - \prod_{i \in U(C)} \left(1 - \min\left(0.99, \max_{d \in C_i} q_d\right)\right), \tag{3}$$

where $U(C)$ indexes distinct source nodes and C_i their detections. The source-wise maximum prevents repeated boxes from multiplying support; the noisy-OR then accumulates evidence from different observing nodes. Clusters with at least two sources or a maximum confidence of at least 0.92 form the primary index $\mathcal{H}^F$. Each is represented by its highest-confidence reachable standoff.

The primary index compresses semantic identity, while the linked residual bank $\mathcal{H}^R$ retains pre-fusion approaches. These access alternatives remain available when a representative is inaccessible. Farthest-first sampling anchored at the representative retains at most B alternatives per cluster; a reserve keeps up to G spatially separated discarded detections per scene/category. Thus B controls within-instance

approach diversity, while G retains hypotheses omitted by cluster filtering. A fixed visit budget prioritizes primary candidates and fills remaining slots with spatially distinct residuals. Both layers count toward storage. Supplementary Sec. A.4 and Algorithm 2 specify association, prototype selection, and visit ordering.

3.3. Multimodal Instance Selection

Category, description, and photograph goals share calibrated survey observations and approach construction; their query evidence determines which instances are selected. For text goals, whole-view RN50 similarity retrieves 30 source nodes; whole-view and local-crop matches select at most two views per node. Grounding DINO [15] localizes regions using the category text or description's first target-bearing sentence. Region d receives $a_d = \langle \mathbf{f}(b_d), \mathbf{g}_q \rangle \sqrt{q_d}$ from its normalized crop feature, normalized complete-query embedding, and detector confidence. In descending a_d order, a depth-grounded region joins the first anchor within 0.75 m horizontally and 0.30 m in source-base height, or creates a new group. Anchors remain fixed, avoiding transitive chains through adjacent objects. Each source contributes once:

$$E(C) = \left[\sum_{i \in U(C)} \left(\max_{d \in C_i} \max(0, a_d) \right)^2 \right]^{1/2}. \qquad (4)$$

This query-conditioned evidence differs from category confidence in Eq. 3. It combines consistent support without letting overlapping crops at one pose dominate. Groups with at least two source nodes are eligible when any exist; otherwise single-source groups remain eligible. One strongest region per source supplies the coordinate-wise median surface estimate. Relevance ordering then retains at most three distinct hypotheses.

For categories, Qwen3-VL-Reranker-2B [13] scores the public text against each group's strongest observed scene-and-crop document. Higher native relevance ranks first; $E(C)$ breaks ties. The reranker orders measured hypotheses without producing coordinates. Descriptions use a fixed parser to ground target and reference phrases separately. Distinct same-frame observations provide near-relation context; depth-quartile world-height intervals identify above/below contradictions. A complete group has usable observations for every parsed target relation and no unsupported predicates. Complete groups are ordered lexicographically by

$$\big(n_{\mathrm{contra}}(C),\ r_{\mathrm{ctx}}(C) + r_{\mathrm{tgt}}(C),\ r_{\mathrm{ctx}}(C)\big), \qquad (5)$$

where n_{contra} counts observed contradictions, r_{ctx} ranks full-query evidence E, and r_{tgt} ranks target-only reranker relevance. Groups with fewer spatial contradictions rank first; rank fusion then balances contextual identity and target appearance. Ranks are computed within the complete subset, which permutes only its existing slots; unresolved groups retain their positions. Fewer than two complete groups preserve the original evidence order.

Photographs query a scene-wide SigLIP2 index [27]. The maximum normalized image cosine over four views and nine fixed crops ranks each node; the ten highest-scoring nodes form appearance hypotheses. Within the best view, crop similarities s_j weight crop centers $\mathbf{u}_j$:

$$\widetilde{\mathbf{u}} = \sum_{j=1}^{9} \frac{\exp(\beta s_j)}{\sum_{\ell=1}^{9} \exp(\beta s_\ell)} \mathbf{u}_j, \qquad \beta = 100. \qquad (6)$$

This pixel determines measured depth and standoff. The reranker compares the actual goal photograph with each original scene-and-crop document; depth-valid hypotheses rank first, then native image relevance, then appearance rank. Scene-wide appearance retrieval can recover a source omitted by the text shortlist, while relevance reranking preserves its measured geometry. All encoders, detector, and reranker remain frozen. Supplementary Sec. A.3 and Algorithm 1 provide complete selection and fallback rules.

3.4. Persistent Execution and Reuse

Each text instance owns an ordered approach block. Its strongest region per source defines a source order by decreasing a_d. For every source, Eq. 2 supplies a desired position from the shared median surface, paired with two snapping options: unrestricted nearest navigable point, then nearest on the source's navigable island. All these source-facing standoffs precede all supporting observation poses in the same source order. Each photograph hypothesis instead retains its own standoff, source-island alternative, and source pose. Concatenating complete blocks preserves semantic priority while retaining distinct access sides. If text grounding yields no valid instance, the direct crop-localization sequence supplies the fallback.

The desired positions and their order are frozen before query-time navmesh access. The controller applies the snapping options in order and skips unreachable approaches. A follower failure allows one same-endpoint replan before advancing. GOAT-Bench's full-navmesh discrete follower issues subtask STOP at its 0.20 m arrival radius; target annotations and evaluator success enter post-hoc scoring. Route feasibility therefore selects among stored approaches without reranking identities. The next request starts at the actual endpoint, reusing survey descriptors from the new pose. Exact query-conditioned detector and relevance responses are cached for repeated query–evidence pairs. Supplementary Sec. A.5 specifies execution and exhaustion behavior.

The HM3D interface additionally permits RGB-D arrival verification to advance after rejection. Survey timing varies along one continuous walk: a policy acquires a prefix before requests and resumes its saved cursor when needed. Its

amortized motion is

$$\overline{D}_N = \frac{D_{\text{survey}} + D_{\text{transfer}} + \sum_{t=1}^{N} p_t}{N}, \qquad (7)$$

where cursor transfers and every executed query leg p_t are charged. This exposes how reusable evidence trades initial acquisition against subsequent navigation.

Online acquisition. On MultiON [28], the same identity–approach principle operates from an initially empty memory. Paid RGB-D observations with episodic GPS and compass build cylinder landmarks, traversed keyframe connectivity, and an observed-free grid. Remembered supports persist across ordered goals. Graph routing selects supporting viewpoints, local grid planning executes their approaches, and compatible multi-view geometry supports conservative goal-region or landmark-based FOUND decisions. A frozen RobustNav RGB-D PointNav policy [4] steers reactive frontiers; observed-map and semantic routes retain geometric execution. The supplement specifies this task-specific online realization.

4. Experiments

GOAT-Bench tests whether one survey can serve successive multimodal requests, while online MultiON tests reuse of observations acquired during navigation. Matched HM3D studies isolate endpoint geometry and memory capacity, and a robot demonstration connects these components to physical execution.

4.1. Benchmarks and Evaluation Setup

GOAT-Bench. We evaluated all 360 Seen, 340 Seen-Synonyms, and 360 Unseen GOAT-Bench episodes, covering category, description, and photograph goals [9]. The splits share validation environments and vary category familiarity. A goal-independent 6,642-node RGB-D survey supplies the persistent memory. Execution uses the released 5,000-action budget, strict 0.25 m geodesic viewpoint criterion, and full-navmesh discrete follower. The query interface receives the released goal photograph for image requests. Table 1 summarizes external paper-reported results, with execution access and evaluator settings described in the caption.

MultiON. MultiON [28] uses online RGB-D with episodic GPS/compass, a 2,500-action budget, and the released 1.5 m FOUND criterion. Policy development used 50 validation episodes per task, followed by confirmation on 100 further validation episodes per task. We then froze the policy and evaluated the first 1,000 released test episodes for each of 1ON, 2ON, and 3ON across 18 Matterport3D scenes.

Controlled studies. The HM3D-Semantics v0.2 validation set contains 1,000 ObjectNav episodes over 36 scenes and six categories [22, 29, 30]. We used Habitat-Sim [25], a goal-independent 0.8 m-cover survey, and GPT-5.4 semantic observations [20]. Endpoint studies use the official 0.1 m goal region; grounding interventions fix detections and rankings, while memory comparisons share source observations, starts, and evaluation. Paired 95% intervals use 10,000 scene-clustered bootstrap samples. Further implementation settings appear in the supplement.

4.2. Benchmark Results

GOAT-Bench. SSTG-Nav achieves the highest displayed SR and second-highest SPL across all three GOAT-Bench splits (Table 1). On Unseen, it reaches 64.5% SR and 42.8% SPL under the strict 0.25 m criterion. These values exceed the reported MTU3D [34] results by 17.3 and 15.1 percentage points, respectively. Query evidence selects instances across modalities and category familiarity, while one shared survey retains their measured approaches.

AstraNav-Memory [6] uses compressed visual history and reports results under a 1 m criterion. The supplement additionally scores our fixed action runs at 1 m to characterize success-distance sensitivity.

MultiON. The online instantiation starts with empty memory and retains observed landmarks and connectivity across subgoals. It reaches 71.9%, 50.6%, and 35.7% sequence Success on 1ON, 2ON, and 3ON (Table 2). Within the original MultiON non-oracle comparison, these Success rates rank second on 1ON/2ON and first on 3ON. On 3ON, Success/Progress are 35.7%/53.2%, compared with the reported 27.0%/46.0% of ProjNeuralMap [28]. A goal transition changes the requested landmark while preserving its earlier supporting views and observed routes. The recorded example in the supplement shows this reuse of an incidentally observed target, complementing the pre-survey setting. The supplement also includes later SGoLAM/MOPA reports with their respective evaluation settings [10, 24].

Reuse across multimodal requests. GOAT-Bench changes the goal modality while retaining the agent's reached position, requiring appearance matching, contextual grounding, and reuse across requests. Figure 3 follows one continuous six-goal sequence. The photograph in Q2 selects stored view A, promoting its node from coarse rank 4 to reranked rank 1. After the rack requests Q4–Q5, the description in Q6 combines bed–pillow context with six distinct observation sources to select view B and return to the same bed. The two reads retain different measured approaches to that target. For Q3, a stored alternative remains executable at Q2's arrival after earlier approaches prove inaccessible, allowing STOP without further motion.

Table 1. GOAT-Bench SR/SPL (%). Baselines are paper-reported results; our evaluation uses all released episodes and the strict 0.25 m viewpoint criterion. AstraNav-Memory reports a 1 m criterion; TANGO and MTU3D use paper-specific or unspecified evaluator settings. MTU3D and SSTG-Nav use simulator-geometry-assisted execution; the other methods use their native controllers. Survey acquisition is analyzed separately from query execution. Bold/underline mark the highest/second-highest displayed value per column across all methods.

Method	Inputs / learned resources	Seen		Seen-Synonyms		Unseen	
		SR	SPL	SR	SPL	SR	SPL
Modular GOAT [2, 9]	RGB-D mapping	26.3	17.5	33.8	24.4	24.9	17.2
Modular CLIP on Wheels [5, 9]	Visual-semantic map	14.8	8.7	18.5	11.5	16.1	10.4
SenseAct-NN Skill Chain [9]	RGB + localization/goal	29.2	12.8	38.2	15.2	29.5	11.3
SenseAct-NN Monolithic [9]	RGB + localization/goal	16.8	9.4	18.5	10.1	12.3	6.8
TANGO [35]	RGB-D + PointGoal	–	–	–	–	32.1	16.5
MTU3D [34]	RGB-D + pose; >1M trajectories	52.2	30.5	48.4	30.3	47.2	27.7
AstraNav-Memory [6]	RGB + pose; 1.5M training samples	<u>65.5</u>	**49.0**	<u>66.8</u>	**54.7**	<u>62.7</u>	**56.9**
SSTG-Nav (ours)	RGB-D memory + pose; goal RGB; frozen encoders + 2B reranker	**66.4**	<u>45.5</u>	**68.7**	<u>44.7</u>	**64.5**	<u>42.8</u>

Table 2. Online MultiON results (%). S/P denote sequence Success/Progress; SPL/PPL are their path-weighted counterparts. Baselines are reported by Wani et al. [28]; their exact episode identities are unreported. Our frozen RGB-D/GPS/compass policy uses 1,000 released test episodes per task, 2,500 actions, and the 1.5 m FOUND criterion. Bold/underline mark the highest/second-highest values, including ties. Later reports and their evaluation settings appear in the supplement.

Method	1ON				2ON				3ON			
	S	P	SPL	PPL	S	P	SPL	PPL	S	P	SPL	PPL
NoMap(RNN) [28]	62.0	62.0	35.0	35.0	24.0	39.0	13.0	21.0	10.0	24.0	4.0	14.0
FRMQN [28]	62.0	62.0	50.0	50.0	29.0	42.0	24.0	33.0	13.0	29.0	11.0	24.0
SMT [28]	63.0	63.0	48.0	48.0	28.0	44.0	26.0	36.0	9.0	22.0	7.0	18.0
EgoMap [28]	69.0	69.0	49.0	49.0	46.0	59.0	<u>31.0</u>	<u>42.0</u>	26.0	44.0	**18.0**	<u>30.0</u>
ProjNeuralMap [28]	70.0	70.0	<u>51.0</u>	<u>51.0</u>	45.0	57.0	30.0	39.0	<u>27.0</u>	<u>46.0</u>	**18.0**	**31.0**
ObjRecogMap [28]	**79.0**	**79.0**	**56.0**	**56.0**	**51.0**	**62.0**	**38.0**	**45.0**	22.0	40.0	<u>17.0</u>	<u>30.0</u>
SSTG-Nav (online)	<u>71.9</u>	<u>71.9</u>	33.4	33.4	<u>50.6</u>	<u>61.4</u>	20.2	24.5	**35.7**	**53.2**	14.1	20.3

Table 3. GOAT-Bench component comparison: SR/SPL (%) on all episodes at 0.25 m. Rows share the survey and navmesh controller. Components accumulate through the complete interface; the last row changes ingress scheduling alone. Bold marks the best SR or SPL in each split.

Query-time interface	Seen	Synonyms	Unseen
Direct crop localization	46.07/33.18	45.40/31.93	38.50/27.92
Metric instances + text recall	52.08/36.17	52.79/35.31	50.49/33.28
+ local photo conditioning	55.76/38.10	56.70/38.06	56.57/37.61
+ category relevance	59.71/40.40	59.18/39.01	57.70/37.64
+ scene-wide photo retrieval	64.34/43.25	65.48/42.82	62.37/41.09
+ native image relevance	65.38/44.63	67.52/43.88	63.73/42.22
+ context–morphology ranks	**66.43**/45.45	**68.66**/44.74	**64.53**/42.83
+ ingress order (alternative)	65.60/**46.61**	67.91/**46.11**	64.37/**45.64**

Thus the shared memory supports both immediate endpoint reuse and later retrieval from a different observed side. All six requests complete in 222 query-execution actions; the visualization re-executes frozen semantic plans with the original controller.

4.3. Ablation and Mechanism Analysis

Multimodal query interface. Table 3 traces the cumulative query interface under a shared survey and controller. Metric instances and text recall raise Unseen SR from 38.50% to 50.49%, a gain of 11.99 percentage points. Pooling observations by spatial instance connects retrieval evidence to the target's measured approaches. Subsequent components refine which instance receives the request while preserving its source geometry. The complete interface improves SR over direct crop localization by 20.4–26.0 points across the three splits.

Each row re-executes full sequences, so later starts include the effects of earlier decisions; row differences measure cumulative policy gains. Scene-wide photograph retrieval adds 4.67 points of Unseen SR by searching beyond the initial text-instance shortlist. Relevance reranking can promote a useful stored view once retrieval includes it, and the instance representation connects that evidence to calibrated approaches.

Combining context and target-morphology ranks adds a further 0.80 points of Unseen SR and yields the highest SR

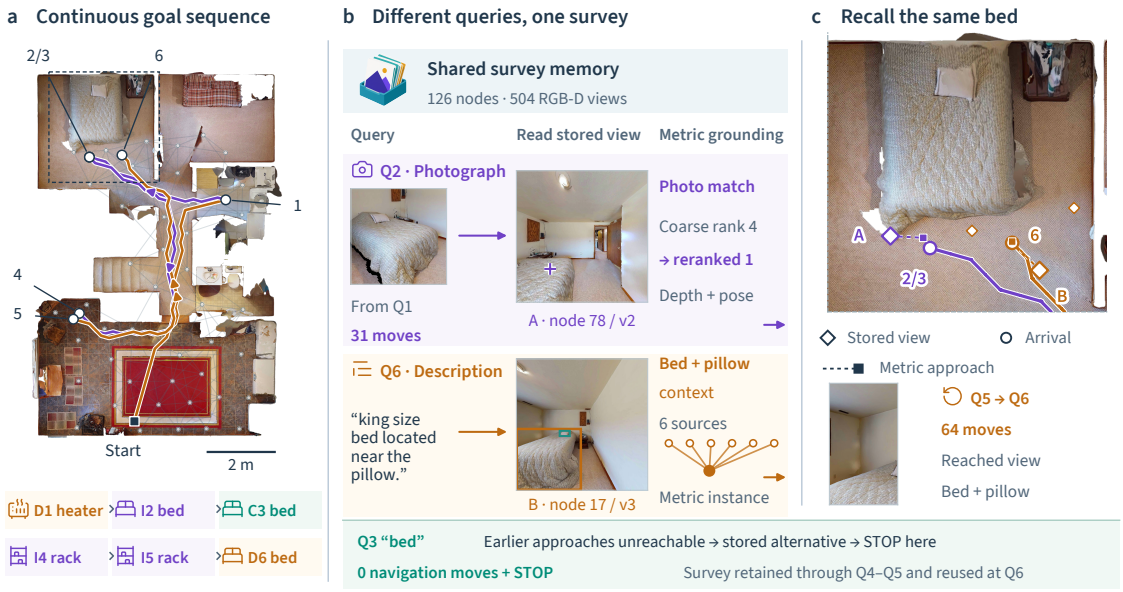


Figure 3. **Different memory reads support a return to the same target.** (a) All six executed paths in one continuous GOAT-Bench episode; D/I/C denote description/image/category goals. Numbers identify actual arrivals; the dashed rectangle locates (c). (b) Q2 matches the goal photograph to stored view A; Q6 uses bed–pillow context and six distinct sources to select B. The cross marks Q2's depth-sampling pixel; the boxes mark Q6's bed and pillow evidence. (c) Diamonds identify stored observation poses, squares selected metric destinations, and circles actual arrivals. Dashed links associate A/B with their destinations; solid lines are executed motion. Q6 returns to Q2's bed after the rack requests. Q3 completes at Q2's arrival with zero navigation moves. All six goals succeed under the strict 0.25 m criterion.

Table 4. Matched HM3D-v2 mechanism studies on 1,000 episodes. (a) Only the endpoint of the frozen Top-1 detection changes. (b) Confidence-ranked primary indices without diversity filtering. (c) Three-candidate capacity with 2 m diversity: up to two primaries then residual fill, or up to three primaries for primary-only; budgeted uses $(B, G) = (8, 4)$. MiB measures serialized primary/residual memory. R@3 and Path@3 are oracle candidate-list recall and path-weighted recall, not autonomous navigation success; compare within each panel.

(a) Metric grounding

Endpoint	SR	SPL
Observation pose	0.252	0.154
Object surface	0.051	0.033
0.8 m standoff	**0.405**	**0.255**

(b) Primary consolidation

Index	Entries	SR	R@3
Raw	20,107	0.405	0.588
Greedy dedup.	3,762	0.406	**0.608**
CC + max	2,236	**0.411**	0.585
CC + noisy-OR	2,236	0.393	0.573
Source-aware	**1,329**	0.388	0.563

(c) Residual capacity

Memory	MiB	R@3	Path@3
Primary only	**1.868**	0.568	0.279
Budgeted	11.385	0.627	**0.288**
Full raw	28.492	**0.634**	**0.288**

in each split. The combined ordering retains both contextual identity and target appearance. An efficiency control changes ingress ordering within each selected text instance while retaining candidate coordinates and cross-instance semantic order. SPL rises by 1.15, 1.38, and 2.81 points on Seen, Synonyms, and Unseen; the paired Unseen gain is 2.81 points [1.53, 4.13]. Preserving alternative approaches therefore makes route efficiency adjustable after semantic selection. The main configuration retains source-evidence ordering.

Approaches turn observations into destinations. On 1,000 HM3D episodes, we fix semantic detections, candidate rankings, and starts, and change only the endpoint of the selected detection (Table 4a). Replacing its observation pose with a reachable 0.8 m standoff raises one-shot SR from 25.2% to 40.5%. The paired gain is 15.3 percentage points [9.4, 21.4], accompanied by a 10.1-point SPL gain [6.3, 14.0]. Direct surface projection achieves 5.1% SR: a visible surface localizes the object but does not supply the required stopping location. The source-facing offset places the endpoint on the observed approach side at the source floor height. Nearby 0.8–1.0 m offsets form a stable operating region (Figure 4a). This intervention isolates the effect of endpoint geometry under fixed semantic evidence.

Compact identity and diverse approaches are complementary. Source-aware fusion reduces the primary index

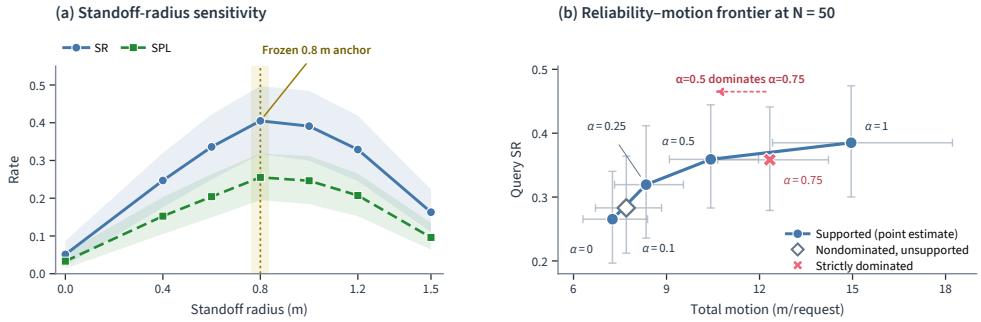


Figure 4. **Geometry and acquisition control navigation.** (a) Standoff sensitivity with semantic evidence and rankings fixed, under the official 0.1 m goal-region criterion. (b) Staged survey budgets at 50 requests, charging query, survey, and saved-cursor transfer motion. Bands and whiskers are scene-clustered 95% intervals. Filled points form the point-estimate lower cost envelope; the hollow point is nondominated but unsupported, and the cross is dominated.

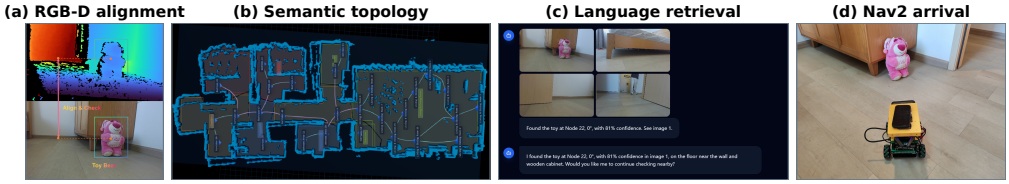


Figure 5. **Physical-system realization.** (a) Aligned RGB-D localizes a toy observation. (b) The survey produces a persistent semantic topology. (c) A natural-language request retrieves Node 22 and its supporting views. (d) The Yahboom X3 reaches the selected target through Nav2. This recorded sequence provides a qualitative demonstration of perception, persistent memory, retrieval, and physical execution.

from 20,107 detections to 1,329 entries, a 93.4% reduction (Table 4b). The paired one-shot SR change is -1.7 points $[-6.4, 3.1]$. Retaining geometry outside this index recovers useful candidate diversity: under the same three-visit budget, the budgeted residual policy raises oracle endpoint recall from 0.568 to 0.627, a paired gain of 0.059 [0.014, 0.104]. The 11.385 MiB budgeted bank retains 98.9% of the full raw bank's oracle recall with 60.0% less serialized storage (Table 4c). This measures the candidate coverage preserved by the memory before arrival decisions.

The source-aware index uses 40.6% fewer entries than Euclidean CC with noisy-OR. Grouping by observation source compresses repeated semantic support; retaining residual geometry protects physically distinct approach options. Among 109 episodes whose first oracle-valid candidate is a budgeted residual, 41 use another approach to an already selected cluster, 53 use another retained cluster, and 15 use the discarded-detection reserve. The capacity benefit therefore spans both another side of a selected instance and hypotheses outside the primary shortlist.

Repeated requests amortize the survey. Changing-start schedules over 43 connected environments measure the cost of reuse, including feasible survey traversal and transfers to resume surveying. At 50 requests, a half-survey captures 78% of the full survey's SR improvement over lazy acquisition with 41% of its additional motion (Figure 4b). The feasible full-survey walk overhead falls from 242.03 m for one request to 4.84 m per request at 50. Thus the reusable representation supports a family of reliability–motion operating points, with survey timing determining how quickly its initial investment becomes useful. Matched schedules vary evidence availability, demonstrating useful reuse before the full survey is complete.

4.4. Physical-System Demonstration

We implement the interface on a Yahboom X3 running ROS 2 Humble [18]. A Gemini 336L supplies RGB-D, an RPLIDAR S2 provides laser scans, and `slam_toolbox` [16] supplies mapping and localization. The persistent manager retains surveyed observations and query state; Nav2 [17] executes the selected metric goal. Figure 5 follows a toy request from calibrated observation and stored topology to retrieval and arrival. This qualitative demonstration integrates calibrated perception, persistent memory, and navigation on a mobile robot. The retrieved observation supplies a metric goal to the robot's mapping and navigation stack, realizing the same observation-to-approach interface used in simulation.

5. Conclusion

SSTG-Nav connects reusable object identity to measured approach geometry, allowing later requests to recover both a target and its observed approach alternatives. A shared survey supports category, description, and photograph goals; a source-aware primary index and residual bank preserve compact retrieval and geometric diversity. GOAT-Bench and online MultiON demonstrate reuse across successive requests, while matched HM3D studies isolate endpoint geometry and quantify memory and acquisition trade-offs. The present evaluation assumes static scenes, with navmesh-assisted execution in the pre-survey benchmarks. Updating memory as scenes change and improving observed-only routing are direct next steps toward persistent deployment.

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SSTG-Nav: Metric-Grounded Spatial-Semantic Topological Graphs for Reusable Object Navigation

Supplementary Material

A. Algorithm and Implementation Details

This supplement specifies the query-to-approach procedure, its category-map realization, and the benchmark execution interfaces. The two realizations share calibrated source observations but use different instance scores: GOAT-Bench groups query-conditioned regions, whereas the HM3D controls consolidate a goal-independent category map. Algorithms S1 and S2 give their respective procedures.

A.1. Survey and Reusable Observations

A calibrated RGB-D survey is constructed independently of future requests. For each HM3D scene, we sample 12,000 navigable proposals with a fixed seed and apply greedy farthest-point selection until the empirical cover radius is at most 0.8 m. Each node considers its eight Euclidean-nearest neighbors; an edge requires Euclidean distance at most 2.4 m and a navigable route at most 3.84 m. Its weight is the route length. The resulting survey contains 6,642 nodes and 21,845 edges across 36 scenes, with four cardinal RGB-D views per node. The same survey underlies the GOAT-Bench interface and HM3D mechanism studies. Disconnected navigable components retain separate connectivity.

Each observation retains its original RGB, depth, intrinsics, camera orientation, and source-agent pose. Normalized whole-view and nine-crop descriptors are computed once. Crops cover half the image in each dimension, with top-left offsets $\{0, 0.25, 0.5\}^2$. The graph organizes this persistent evidence and survey acquisition; the formal GOAT controller uses the released navmesh planner for query execution. Repeated requests reuse the survey descriptors and exact cached query–observation responses, while the current pose changes after every executed subtask.

A.2. Calibrated Grounding and Approach Geometry

Habitat depth is camera-forward depth, rather than Euclidean range along a pixel ray. We use camera coordinates $+x$ right, $+y$ up, and $-z$ forward. For normalized pixel coordinates (u, v), measured forward depth z, horizontal field of view ϕ_h, and image size $W \times H$,

$$\phi_v = 2\tan^{-1}\left(\frac{H}{W}\tan\frac{\phi_h}{2}\right), \tag{S1}$$

$$\mathbf{x}_c = \left(2(u-\tfrac{1}{2})z\tan\tfrac{\phi_h}{2}, -2(v-\tfrac{1}{2})z\tan\tfrac{\phi_v}{2}, -z\right). \tag{S2}$$

Table S1. Principal fixed parameters. HM3D fusion and GOAT-Bench instance association operate at different representation levels.

Parameter	Value
Survey RGB-D / optical height	640×640 / 1.25 m
Survey horizontal / vertical FoV	120° / 120°
Metric standoff / depth patch	0.8 m / 7×7
Valid forward-depth interval	$(0.2, 6.0)$ m
HM3D fusion: horizontal / vertical	1.2 m / 1.0 m
HM3D fusion: geodesic / singleton	3.0 m / confidence 0.92
GOAT association: horizontal / floor	0.75 m / 0.30 m
GOAT detector: box / text threshold	0.25 / 0.25
GOAT text nodes / instance blocks	30 / at most 3
GOAT image nodes / crop softmax	10 / $\beta = 100$
GOAT controller arrival radius	0.20 m
MultiON grid / keyframe translation	0.25 m / 0.60 m

The camera's calibrated world transform maps $\mathbf{x}_c$ to a surface estimate $\widehat{\mathbf{o}}$. This transform includes the optical offset above the source agent's base pose $\mathbf{p}_i$; the base and camera positions are therefore distinct. For text regions, (u, v) is the detected box center. For photographs, it is the soft crop center defined below.

We take the median finite depth in a 7×7 patch, retaining values strictly between 0.2 and 6.0 m. A text-region patch is additionally clipped to its bounding box. With no valid sample, the region supplies no measured surface. A horizontal source-facing offset of 0.8 m from $\widehat{\mathbf{o}}$ supplies the desired standoff, with its vertical coordinate set to the source agent's base height. For GOAT text instances, a source within 0.8 m of the estimated surface retains its own pose rather than extrapolating behind the camera. Photograph localization uses the original fixed-radius construction and omits the standoff when the horizontal source ray is shorter than 0.2 m. Its stored observation pose remains available even when depth or the ray is invalid.

Desired positions are measured hypotheses, not automatically executable destinations. The HM3D category-map construction retains a snapped standoff only when it has a navigable connection to its source. GOAT instead stores both an unrestricted nearest-navmesh snap option and an option constrained to the source's navigable island, then checks reachability from the current pose during execution. This distinction preserves source-supported alternatives without treating a geometrically nearby point on another floor as interchangeable with the source observation.

A.3. GOAT-Bench Multimodal Query Procedure

Text retrieval and region evidence. For the normalized released CLIP ResNet-50 (RN50) goal embedding $\mathbf{g}_q$, the score of node i is the maximum whole-view cosine $\max_k \langle \mathbf{f}_{ik}, \mathbf{g}_q \rangle$. The highest-scoring 30 nodes are retrieved, with node ID breaking exact score ties. At each node we retain the best whole view and the view containing the best local crop, merging them when they coincide. Thus at most two original views per node enter query-conditioned grounding. Frozen Grounding DINO [15] uses the category text or description's first target-bearing sentence, with box and text thresholds of 0.25. It retains at most five confidence-ranked boxes per view; boxes smaller than four pixels in either dimension are discarded. A 5% padded region is encoded by RN50. The region score is

$$a_d = \langle \mathbf{f}_d, \mathbf{g}_q \rangle \sqrt{q_d}, \tag{S3}$$

where q_d is detector confidence and $\mathbf{f}_d$ is the normalized region descriptor. Geometry uses the original box center and calibrated depth, rather than the padded crop's center.

Depth-valid regions are processed in descending a_d. Each joins the first existing anchor whose surface estimate is within 0.75 m horizontally and whose source base height differs by at most 0.30 m, or starts a new group. Comparing to a fixed anchor avoids a transitive chain across adjacent objects. Each source node contributes its strongest region once to both the coordinate-wise median surface estimate and the evidence energy

$$E(C) = \left[\sum_{i \in U(C)} \left(\max_{d \in C_i} \max(0, a_d) \right)^2 \right]^{1/2}. \tag{S4}$$

Here $U(C)$ contains the distinct source nodes supporting C, and C_i contains their regions. Taking one maximum per node prevents overlapping crops from inflating support. If any group has at least two source nodes, only such groups are eligible; otherwise all groups are retained. The original context order sorts eligible groups by decreasing $E(C)$, decreasing strongest-region score, and increasing best retrieval rank, with stable group order resolving remaining ties. If no region yields valid geometry, the original whole-view/crop localizer supplies the fallback candidates.

Category relevance. For each eligible group, its strongest original region supplies a visual document: a 640×640 scene image paired with a 320×640 magnified crop, using 5% padding. Qwen3-VL-Reranker-2B [13] scores the public category text against these observed documents. Its native relevance orders groups; ties use $E(C)$, strongest-region score, best retrieval rank, and group order. At most three groups are selected. Relevance changes identity ordering while each group's measured surface and complete approach block stay fixed.

Description roles and ordinal fusion. A fixed parser uses public category vocabulary and explicit sentence subjects to separate target appearance clauses from target–reference relations. Ambiguous subjects or unresolved directives retain the original policy decision. Target and reference phrases are grounded independently in the same retrieved views; these auxiliary detections supply relation evidence but never create navigation coordinates. A target role matches an original instance only within its source frame and the same 0.75 m/0.30 m association gates.

A target–reference pair must have finite depth support in the same frame. Box IoU at least 0.95 together with overlapping depth interquartile intervals identifies a repeated measurement of one surface and supplies no relation evidence. Distinct co-observed regions supply contextual support for *near*, *next to*, *beside*, and *close to*; this is co-occurrence evidence, without a fitted metric near-distance threshold. For *above/below*, backprojecting the depth quartiles gives world-height intervals $[y_t^-, y_t^+]$ and $[y_r^-, y_r^+]$. For example, *above* is supported when $y_t^- > y_r^+$ and contradicted when $y_t^+ < y_r^-$; overlapping intervals remain unknown. Unsupported predicates and reference-to-reference relations remain unknown.

A group is complete only when every parsed target relation has usable evidence and no unsupported relation remains. An edge is contradictory only when all its usable observed pairs contradict it. With at least two complete groups, let r_c be rank in their original full-query context order and r_t rank by native relevance to target-only appearance text. Complete groups sort lexicographically by

$$\big(n_{\text{contradictions}},\ r_c + r_t,\ r_c\big). \tag{S5}$$

Only their occupied slots are permuted; incomplete groups retain their exact original slots. Fewer than two complete groups retain the original decision. This rule combines explicit geometry, contextual identity, and target appearance without allowing absent relation evidence to act as a negative label.

Photograph retrieval and localization. A trusted adapter renders the photograph selected by the released image-goal sensor and restores agent and sensor state. Only copied RGB pixels enter the policy. SigLIP2 B/16-224 [27] compares this photograph with the nine fixed overlapping crops in each of the four original views at every scene node. For crop similarity s_{ikr}, node score is $\max_{k,r} s_{ikr}$; the ten highest-scoring nodes are kept, with smaller node ID resolving ties. The maximum crop selects view k^*, and the nine similarities in that view give the depth-sampling pixel

$$\widetilde{\mathbf{u}}_i = \sum_{r=1}^{9} \frac{\exp(100(s_{ik^*r} - s_i^{\max}))}{\sum_{r'} \exp(100(s_{ik^*r'} - s_i^{\max}))} \mathbf{c}_r, \tag{S6}$$

where $\mathbf{c}_r$ is the normalized crop center and $s_i^{\max} = \max_{k,r} s_{ikr}$. Subtracting this maximum stabilizes softmax computation. Qwen3-VL-Reranker-2B scores the goal photograph against scene-and-crop documents at these ten nodes. Complete node blocks are sorted first by valid depth, then decreasing native image relevance, then original SigLIP rank and node ID. Reranking preserves the chosen pixel, depth, and coordinates. Invalid-depth nodes retain their measured source poses as fallbacks.

In Algorithm S1, $\|$ concatenates complete approach blocks, and d_i^* is the strongest region from source i. RANKCATEGORY, RANKDESCRIPTION, and RANKPHOTO implement the ordering rules above. INDEPENDENTELIGIBLE prefers groups with two distinct sources when any exist. PREPARE performs snapping and reachability tests; FOLLOW emits actions until arrival, follower failure, or the budget guard. The guard reserves one STOP for each of the n_{rem} remaining subtasks, including the current request. Section A.5 specifies approach blocks and stopping behavior. Cache updates retain query-conditioned responses for reuse.

A.4. HM3D Category Consolidation and Residual Memory

For fixed-response HM3D studies, GPT-5.4 [20] receives four indexed original views and returns category, normalized bounding box, view index, and confidence for chair, bed, plant, toilet, TV/monitor, and sofa. One shared detection set supplies every endpoint representation. Invalid depth, a source ray below 0.2 m, failed snapping, or no source-to-standoff path excludes that detection from the metric candidate bank.

Within each category, two detections are linked when their estimated surfaces differ by at most 1.2 m horizontally and 1.0 m vertically, and their snapped standoffs have a finite geodesic separation at most 3.0 m. Union-find connected components define clusters. This category-map association differs from the non-transitive query-region anchors above: it groups a fixed scene-wide detection bank. Source-connected approach geometry constrains association between spatially close but geodesically separated regions. A cluster is retained when it has at least two distinct source nodes or its maximum confidence is at least 0.92. Each node contributes its maximum confidence once to

$$Q(C) = 1 - \prod_{i \in U(C)} \left(1 - \min(0.99, \max_{d \in C_i} q_d)\right). \quad \text{(S7)}$$

The highest-confidence member represents the cluster in the primary index; original detection order resolves a confidence tie. Thus $Q(C)$ ranks category evidence, whereas $E(C)$ ranks query-conditioned region support. They are used in their respective experiments, rather than combined into a shared calibrated probability.

Algorithm S1 GOAT query-to-action procedure

Require: Survey observations $\mathcal{O}$, features $\mathcal{F}$, cache $\mathcal{M}$, query q, pose $\mathbf{p}$, action budget L, remaining subtasks n_{rem}
Ensure: Executed actions, realized pose $\mathbf{p}$, updated cache $\mathcal{M}$

if q.type $\in \{\text{category}, \text{description}\}$ **then**
  $N \leftarrow \text{Top}_{30}(\max_k \langle \mathbf{f}_{ik}, \mathbf{g}_q \rangle)$
  $V \leftarrow$ BESTVIEWS(N); $D \leftarrow$ GROUND(q, V)
  $a_d \leftarrow \langle \mathbf{f}_d, \mathbf{g}_q \rangle \sqrt{q_d}, \forall d \in D$
  $D \leftarrow$ LOCALIZE$(D, \mathcal{O})$
  $\mathcal{C} \leftarrow$ ANCHORGROUPS$(D, 0.75, 0.30)$
  if $\mathcal{C} = \varnothing$ **then**
    $\mathcal{A} \leftarrow$ COARSECROPAPPROACHES$(q, N, \mathcal{O})$
  else
    $\mathcal{C} \leftarrow$ INDEPENDENTELIGIBLE$(\mathcal{C})$
    $\mathbf{o}_C \leftarrow \text{median}\{\mathbf{o}_{d_i^*} : i \in U(C)\}, \forall C$
    $E_C \leftarrow \sqrt{\sum_{i \in U(C)} (\max_{d \in C_i} \max(0, a_d))^2}, \forall C$
    if q.type $=$ category **then**
      $\pi \leftarrow$ RANKCATEGORY$(\mathcal{C}, q, E)$
    else
      $\pi \leftarrow$ RANKDESCRIPTION$(\mathcal{C}, q, E)$
    end if
    $\mathcal{A} \leftarrow \|_{C \in \pi[1:3]}$ TEXTBLOCK$(C, \mathbf{o}_C)$
  end if
else
  $s_{ikr} \leftarrow \langle \mathbf{f}_{ikr}^{\text{SigLIP}}, \mathbf{g}_q^{\text{image}} \rangle$
  $N \leftarrow \text{Top}_{10}(\max_{k,r} s_{ikr})$
  $P \leftarrow$ SOFTCROPPIXELS$(N, s, 100)$
  $D \leftarrow$ LOCALIZE$(P, \mathcal{O})$
  $\pi \leftarrow$ RANKPHOTO(N, D, q)
  $\mathcal{A} \leftarrow \|_{i \in \pi}$ PHOTOBLOCK(i, D_i)
end if
$\mathcal{M} \leftarrow$ CACHERESPONSES$(\mathcal{M}, q)$; $\mathcal{A} \leftarrow$ FREEZE$(\mathcal{A})$
for $a \in \mathcal{A}$ **do**
  for $r \in \{0, 1\}$ **do** ▷ At most one same-endpoint replan
    $\gamma \leftarrow$ PREPARE$(a, \mathbf{p})$
    if $\gamma = \varnothing$ **then**
      break
    end if
    $(\mathbf{p}, \sigma, L) \leftarrow$ FOLLOW$(\gamma, 0.20, L, n_{\text{rem}})$
    if $\sigma \in \{\text{arrived}, \text{budget_guard}\}$ **then**
      SUBTASKSTOP; **return** $(\mathbf{p}, \mathcal{M})$
    end if
    if $\sigma \neq \text{follower_error}$ **then**
      break
    end if
  end for
end for
SUBTASKSTOP; **return** $(\mathbf{p}, \mathcal{M})$

For a residual budget (B, G), the representative anchors farthest-first selection of at most B additional members, maximizing each candidate's minimum Euclidean standoff distance to the representative and already retained alternatives. Ties prefer higher confidence and then stable source-view/detection identity. A second reserve scans detections

excluded by cluster retention in decreasing confidence, preserving at most G per scene/category at least 2 m from already retained approaches and earlier reserve members. This construction uses no requests, episode starts, or valid goal viewpoints. The selected $(8, 4)$ bank retains multiple observed access sides and a small reserve of otherwise omitted hypotheses.

The reported $K = 3$ residual diagnostic first takes up to two reachable, 2 m-separated category primaries, then fills remaining slots from the residual bank under the same separation rule. Primaries rank by $Q(C)$ then start-to-endpoint geodesic distance; residuals rank by original confidence, cluster support, and distance, with stable IDs resolving ties. The primary-only reference uses up to three primaries. Recall@3 and Path@3 score the resulting frozen lists with evaluator feedback; they measure candidate capacity. These diagnostic lists do not define the GOAT controller's STOP decisions.

In Algorithm S2, COMPATIBLE applies the 1.2 m horizontal, 1.0 m vertical, and 3.0 m geodesic gates above; D_{ret} contains every member of a retained cluster. MINSEPARATION uses Euclidean standoff distance and returns infinity for an empty anchor set. Farthest-first ties prefer higher confidence and stable evidence identity; primary ties preserve detection order. These conventions make the budgeted construction deterministic.

Algorithm S2 Source-aware category memory construction

Require: Single-scene detections D_0, calibrated RGB-D/poses $\mathcal{O}$, per-cluster budget B, per-category reserve G
Ensure: Primary index $\mathcal{H}^F$ and residual bank $\mathcal{H}^R$

$D \leftarrow \varnothing$; $\mathcal{H}^F \leftarrow \varnothing$; $\mathcal{H}^R \leftarrow \varnothing$; $D_{\text{ret}} \leftarrow \varnothing$
for $d \in D_0$ **do**
  $(\mathbf{o}_d, \mathbf{s}_d) \leftarrow \text{GROUNDSTANDOFF}(d, \mathcal{O}, 0.8)$
  if $\text{VALIDDEPTHRAY}(d) \wedge \text{SOURCECONNECTED}(\mathbf{s}_d, d)$ **then**
    $D \leftarrow D \cup \{d\}$
  end if
end for
$\text{UF} \leftarrow \text{MAKESETS}(D)$
for $(d_i, d_j) \in D \times D$ with $i < j$ **do**
  if $c_i = c_j \wedge \text{COMPATIBLE}(d_i, d_j)$ **then**
    $\text{UNION}(\text{UF}, d_i, d_j)$
  end if
end for
for $C \in \text{COMPONENTS}(\text{UF})$ **do**
  $d^* \leftarrow \arg\max_{d \in C} q_d$
  if $|U(C)| \geq 2 \vee q_{d^*} \geq 0.92$ **then**
    $Q_C \leftarrow 1 - \prod_{i \in U(C)} (1 - \min(0.99, \max_{d \in C_i} q_d))$
    $\mathcal{H}^F \leftarrow \mathcal{H}^F \cup \{(d^*, Q_C, C)\}$; $D_{\text{ret}} \leftarrow D_{\text{ret}} \cup C$
    $A_C \leftarrow \{d^*\}$
    while $|A_C| < B + 1 \wedge C \setminus A_C \neq \varnothing$ **do**
      $d \leftarrow \arg\max_{x \in C \setminus A_C} \min_{a \in A_C} \|\mathbf{s}_x - \mathbf{s}_a\|_2$
      $A_C \leftarrow A_C \cup \{d\}$; $\mathcal{H}^R \leftarrow \mathcal{H}^R \cup \{d\}$
    end while
  end if
end for
for $c \in \text{CATEGORIES}(D)$ **do**
  $S_c \leftarrow \varnothing$
  for $d \in \text{SORTCONFIDENCE}(D_c \setminus D_{\text{ret}})$ **do**
    if $|S_c| = G$ **then**
      break
    end if
    if $\text{MINSEPARATION}(d, \mathcal{H}^F_c \cup \mathcal{H}^R_c \cup S_c) \geq 2$ **then**
      $S_c \leftarrow S_c \cup \{d\}$
    end if
  end for
  $\mathcal{H}^R \leftarrow \mathcal{H}^R \cup S_c$
end for
return $(\mathcal{H}^F, \mathcal{H}^R)$

A.5. Approach Blocks, Execution, and Stopping

An approach block contains the ordered destinations associated with one instance. For each GOAT text instance, sources follow decreasing region score, each supplying an unrestricted standoff snap option and then its source-island-constrained option. Supporting observation poses follow these object-relative standoffs as fallbacks. Every approach to one selected instance precedes the next instance's block. A photograph node instead supplies its unrestricted standoff, its source-island-constrained alternative, and its recorded observation pose, omitting invalid standoffs. The semantic ordering and desired endpoints are fixed for the current request before controller geometry access.

The formal GOAT controller snaps the next approach according to its stored option, rejects non-finite positions and unreachable routes, and uses the full-navmesh `ShortestPathFollower` to emit official discrete actions. A follower failure permits one replan to the same endpoint from the updated pose; persistent failure advances to the next frozen approach. Reaching the 0.20 m controller radius ends the query's navigation loop and issues subtask STOP. Exhausting the finite candidate sequence also issues subtask STOP at the actual pose. A budget guard reserves one STOP action for every remaining subtask. The next request starts at the realized pose, and all executed actions count toward query SPL. Reaching a controller endpoint does not expose evaluator acceptance to the policy: target coordinates, valid goal viewpoints, target distance, and success are used only for post-hoc scoring.

A separate HM3D arrival-verification realization captures four fresh 120° RGB-D views at a reached candidate, with view zero facing the stored object estimate. GPT-5.4 receives these views and a boxed mapping-time reference. Acceptance requires a visible requested target, a valid stopping side, confidence at least 0.75, and median aligned depth in $[0.25, 2.5]$ m. Rejection continues from the current pose to the next candidate in a raw 3 m-separated list. These

verifier-controlled lists and the 2 m primary/residual capacity diagnostic are separate protocols; the GOAT main results use the controller-arrival rule above.

Survey features persist across requests; query-conditioned detector and relevance responses are reused for identical query–evidence pairs. Encoders, detector, and reranker use their released pretrained weights without GOAT-specific fine-tuning. Query inference remains distinct from survey acquisition and feature encoding, and official query-action SPL measures navigation after the survey.

A.6. Online MultiON Instantiation

MultiON [28] starts every episode with an empty graph, landmark bank, and occupancy grid. The policy receives RGB-D, current goal category, episodic GPS and compass, and action acknowledgements. Keyframes are added after 0.60 m translation; paid rotations extend their observations. Traversed edges and observed-free loop closures within 1.50 m connect keyframes. Depth-projected cylinder observations persist across subgoals. Dijkstra routes connect supporting viewpoints; eight-connected grid search supplies local observed-free paths. Exploration over the observed grid selects a central cell in the largest connected frontier component with a bounded 50-action horizon. A frozen RobustNav RGB-D PointNav policy [4] steers the current unblocked reactive frontier only when no map route or semantic landmark key is active. Its inputs are current RGB-D and the frontier's compass-aligned polar displacement; observed-map and semantic routes retain geometric execution. Each reactive frontier has 50 cumulative paid steering actions across interruptions and goal changes, with at least 0.0625 m progress per eight consecutive actions. Failure of either bound blocks that frontier and triggers reselection; the next observation confirms each committed action before charging it.

Compatible full-component measurements from at least two keyframes define an observed goal region. For axial span ℓ_v, the center uncertainty is $\epsilon_v = \max(0, (1.5 - \ell_v)/2) + 0.1$ m. A free cell is eligible when its horizontal distance to every supported center, plus ϵ_v and the grid-cell half-diagonal, is below 1.5 m. Region-based FOUND requires a matching current full RGB-D component and an observed corridor to its surface. Otherwise, verified-landmark control uses the original 0.70 m surface gate and bounded paid reacquisition. New spatial evidence can reactivate a previously failed route. Native network STOP or a frontier distance at most 0.42 m invokes geometric steering on the same observation; task-level FOUND retains the evidence gates above. Recurrent state resets on the first network request, a goal or 0.25 m frontier-cell change, or intervening paid observations handled by another controller.

The released sensor is 256×256 RGB-D with a 79° horizontal FoV and 0.88 m optical height. The frozen visual policy uses deterministic modal actions, crops sensor rows 32–223 and resizes them to 224×224 inputs, and applies ImageNet RGB normalization and depth normalization $(d - 0.5)/0.25$. The agent uses 0.25 m forward motion and 30° turns, with a 2,500-action budget. Success follows the released 1.5 m geodesic FOUND condition; an incorrect declaration terminates the episode. Formal evaluation uses the first 1,000 episodes in released test order for each task over 18 scenes. Development uses the first 50 compact-validation episodes per task, followed by confirmation on the next 100 episodes per task before formal testing. These episodes are distinct, with overlapping scenes. MultiON planning uses observed geometry; the evaluator computes ground-truth success distances.

Table S2. GOAT-Bench by query modality under the strict 0.25 m evaluator. SR/SPL are subtask-micro percentages; n counts subtasks.

Split	Modality	n	SR	SPL
Seen	Category	986	59.63	35.50
	Description	881	50.06	37.86
	Photograph	890	89.55	63.79
Seen-Synonyms	Category	881	62.54	38.76
	Description	847	59.50	40.83
	Photograph	797	86.32	55.62
Unseen	Category	991	54.79	31.43
	Description	856	53.86	37.85
	Photograph	822	88.32	61.69

B. Additional Evaluation

B.1. GOAT-Bench Modalities and Thresholds

The complete GOAT-Bench [9] evaluation contains 1,060 episodes and 7,951 ordered subtasks. Seen, Seen-Synonyms, and Unseen contain 360/340/360 episodes across 36/34/36 scenes, respectively. The released strict evaluator uses geodesic distance below 0.25 m to valid goal viewpoints and a 5,000-action episode budget. Main results macro-average official episode partial success and composite SPL. Table S2 instead groups individual subtasks by modality. This breakdown shows how the common metric memory serves category, description, and photograph evidence without changing the task definition.

The same selected policy is also scored at a 1 m acceptance distance in Table S3. Survey, query evidence, semantic order, 0.20 m controller arrival radius, and complete endpoint attempts remain unchanged. The wider tolerance increases acceptance of the same executed approaches. The strict released condition remains the primary benchmark result. Whole-scene intervals use 100,000 resamples with seed 20261012; paired comparisons apply the same sampled scenes to both conditions and retain all their episodes. This

Table S3. GOAT-Bench strict evaluation and success-distance sensitivity. SR/SPL are official episode-macro percentages. Brackets give 95% whole-scene bootstrap intervals. Paired changes are 1 m minus 0.25 m in percentage points; both conditions retain the same endpoint attempts and executed action counts.

	Strict 0.25 m		1 m sensitivity		Paired change [95% CI]	
Split	SR [95% CI]	SPL [95% CI]	SR	SPL	ΔSR	ΔSPL
Seen	66.43 [63.27,69.59]	45.45 [42.62,48.23]	71.44	49.17	+5.01 [3.66,6.50]	+3.71 [2.59,4.99]
Seen-Synonyms	68.66 [62.45,74.64]	44.74 [40.89,48.41]	75.54	49.29	+6.89 [4.41,9.81]	+4.56 [3.01,6.29]
Unseen	64.53 [59.48,69.52]	42.83 [39.36,46.37]	70.11	46.88	+5.58 [3.74,7.61]	+4.05 [2.67,5.59]

Table S4. HM3D standoff sensitivity with frozen detections and ranking. SR/SPL are percentages; intervals describe paired SR changes relative to the stored 0.8 m anchor. Radius zero snaps the estimated object surface.

Radius (m)	SR	SPL	ΔSR [95% CI]
0.0	5.1	3.3	−35.4 [−43.6,−27.1]
0.4	24.7	15.3	−15.8 [−20.8,−10.8]
0.6	33.6	20.4	−6.9 [−10.3,−3.7]
0.8	**40.5**	**25.5**	0.0
1.0	<u>39.1</u>	<u>24.6</u>	−1.4 [−4.8,2.1]
1.2	32.9	20.7	−7.6 [−14.1,−1.5]
1.5	16.3	9.6	−24.2 [−33.3,−15.3]

preserves within-scene dependence and the official episode aggregation.

B.2. Grounding and Memory Controls

HM3D mechanism studies use the 1,000 validation episodes of HM3D-Semantics v0.2 [29, 30] across 36 scenes. Endpoint success is geodesic distance below 0.1 m to released valid viewpoints. The fixed-response experiment preserves all 20,107 depth-valid detections, scores, episode starts, reachable pools, ranking, and selected detection; only its attached endpoint changes. The source observation pose obtains 25.2% SR/15.4% SPL, while a 0.8 m reachable standoff obtains 40.5%/25.5%. The paired improvements are 15.3 [9.4,21.4] SR and 10.1 [6.3,14.0] SPL percentage points under 10,000 whole-scene resamples. Official goal information enters after endpoint selection.

The radius study in Table S4 supports an approach-side offset rather than a direct surface destination. Selecting the radius on 35 scenes and evaluating on the remaining scene chooses 0.8 m in every leave-one-scene-out fold. The out-of-fold result is 40.5% SR/25.5% SPL, providing an internal scene-stability check for this parameter.

Source-aware consolidation reduces the primary index from 20,107 detections to 1,329 representatives. Under independently ranked representations, raw and consolidated Top-1 SR are 40.5% and 38.8%; their paired difference is −1.7 [−6.4,3.1] points. Consolidation therefore provides compact retrieval while the residual bank preserves approach diversity. The primary-index count excludes linked residual standoffs and multimodal appearance features. GOAT-Bench's scene-wide SigLIP2 bank contains 239,112 normalized 768-dimensional FP32 vectors, occupying 700.5 MiB in addition to RN50 features and retained RGB-D. These quantities describe complementary storage layers.

For changing-start reuse, the HM3D study forms 43 scene/component environments and 50 fixed request schedules per environment. Every horizon uses the same schedule prefixes. The first request starts at a released episode pose; each later request starts at the preceding selected endpoint. A continuous component-level open tree walk supplies the survey-cost bound, with saved-cursor transfers charged in staged schedules. At horizon 50, the full-survey raw-standoff policy achieves 38.5% query SR with 14.96 m total motion per request, including 4.84 m survey and 0.56 m transfer motion. The half-survey policy achieves 35.9% with 10.43 m per request. These measurements separate evidence acquisition from query travel under matched observations and request sequences.

B.3. MultiON Uncertainty and Published Context

Table S5 provides scene-level uncertainty for the complete online evaluations and retains later published 3ON reports with their stated settings. Success measures complete ordered navigation, Progress the completed fraction, and SPL/PPL their path-weighted counterparts. Our intervals resample entire scenes 10,000 times, keeping all episodes in each selected scene, with seed 20261012. SGoLAM [10] reports the CVPR 2021 challenge with a 1.5 m threshold and unspecified exact episode inventory. MOPA [24] reports 1,050 MP3D test episodes over 18 scenes at 1 m. Their reported settings remain explicit alongside the results.

C. Qualitative Demonstration

Figure S1 follows an incidental observation under the geometric reference controller across a MultiON goal transition. The pink cylinder is observed while red is still the active goal, then becomes the next request at action 49. Its stored source views and observed connectivity support the approach; a matching current RGB-D observation supports FOUND at

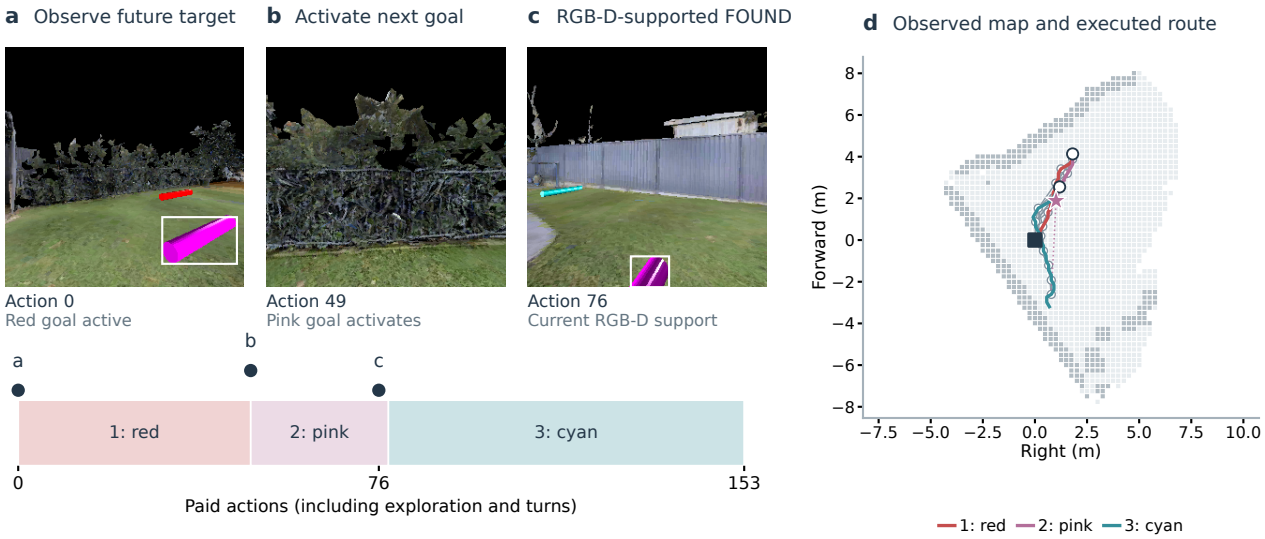


Figure S1. Incidental observation and later reuse in a second MultiON scene. (a) Pink is observed before its request. (b) The current goal changes after red is completed. (c) The final pink declaration is supported by matching current RGB-D. (d) The observed grid, keyframe topology, landmark supports, and executed trajectory connect all three requests. Frames come from simulator action re-execution with the frozen geometric reference controller; the recorded episode result and policy diagnostics match the evaluation record.

Table S5. MultiON complete evaluations and later published 3ON context. All metrics are percentages. Our rows contain 1,000 episodes per task over 18 scenes, with 95% whole-scene intervals. Published rows retain their original evaluation protocols and reported ± terms.

Task / method	Success	Progress	SPL	PPL
SSTG-Nav, released test prefix, 1.5 m, 2,500 actions				
1ON	71.90 [67.97,75.63]	71.90 [67.97,75.63]	33.37 [29.52,37.95]	33.37 [29.52,37.95]
2ON	50.60 [44.46,56.98]	61.35 [55.89,66.88]	20.22 [16.56,24.40]	24.48 [20.97,28.42]
3ON	35.70 [28.34,43.57]	53.20 [47.60,58.90]	14.10 [10.62,18.05]	20.30 [17.21,23.58]
Later published 3ON reports, original settings				
SGoLAM [10], challenge, 1.5 m	62	71	34	39
MOPA (PredSem) [24], MP3D, 1 m	38 ± 2	53 ± 2	15 ± 1	21 ± 1

action 76. Red, pink, and cyan are completed in 153 paid actions. The figure exposes the relationship between evidence acquisition, persistent landmark identity, and subsequent navigation.

The accompanying simulation videos provide first-person observations and synchronized query/memory views. MultiON recordings execute the geometric reference and the selected reactive RGB-D controller. The GOAT-Bench recording re-executes the archived frozen semantic candidate sequence through its navmesh-assisted controller, preserving the six-subtask outcome and action counts. In this continuous sequence, a photograph request reaches a bed, the following category request completes at the same pose, and a later description returns to that bed after two rack requests. These queries read different supporting views from the same calibrated survey; the MultiON case instead illustrates evidence acquired during paid navigation. Each video uses one actual rendered observation per action at 8 fps.